E00 — Smoke test

Known-answer checks of every Regina feature the later experiments rely on

regina_experiments
2026-10-19

Summary

All 15 checks pass with Regina 7.4.1 (the official Ubuntu 24.04 package) under Python 3.12.3. The whole suite runs in under a second. CI runs it on every push, as part of `make test`.

Purpose

Later experiments compute invariants in bulk and compare them across thousands of triangulations. Before trusting those numbers, we check that each part of Regina we depend on returns the right answer on a few inputs whose answer is known independently. These checks also catch an environment problem early, e.g. a Python interpreter that cannot import the `regina` module.

There are two kinds of check:

- **Mathematical facts.** The expected value comes from the literature and is independent of Regina.
- **Regression pins.** The expected value is Regina's own output, recorded so that an upgrade that changes it is noticed. These are marked *(pin)* below.

Checks

#	Check	Expected	Why it is right
1	$H_1(L(7,2))$	$\mathbb{Z}_7$	$\pi_1(L(p,q)) = \mathbb{Z}_p$
2	H_1 of the Poincaré homology sphere	0	homology sphere
3	`isSphere()` on a 3-sphere triangulation	true	—
4	`isSphere()` on the Poincaré sphere	false	$\lvert\pi_1\rvert = 120$
5	summands of $\mathbb{RP}^3 \# \mathbb{RP}^3$	2	prime decomposition
6	census lookup of the figure-eight complement	`m004`	SnapPea cusped census
7	volume of the figure-eight complement	2.0298832128…	two regular ideal tetrahedra, $6\Lambda(\pi/3)$
8	volume of the Weeks manifold	0.9427073627…	smallest closed hyperbolic 3-manifold (Gabai–Meyerhoff–Milley)
9	figure-eight complement has a strict angle structure	true	all angles $\pi/3$

#	Check	Expected	Why it is right
10	vertex normal surfaces of the figure-eight complement (quad coords)	4	*(pin)*
11	$TV_5(S^3) \neq TV_5(L(5,1))$	distinct	*(pin; the values are Regina's normalisation)*
12	Jones polynomial of the right-handed trefoil	$-x^8 + x^6 + x^2$	$x = \sqrt{t}$, i.e. $t + t^3 - t^4$; sources differ by the mirror $t \mapsto t^{-1}$
13	complement of the figure-eight knot (from a diagram)	`m004`	as check 6, via `Link.complement()`
14	$\chi(\mathbb{CP}^2)$	3	$1 + 1 + 1$
15	intersection form of K3	rank 22, signature −16, even	$2(-E_8) \oplus 3H$

The volumes in checks 7–8 must match to 10^{-8}. Every other check is an exact comparison.

Results

Output of the run on 2026-10-19 (git `b62487a`, 4-core cloud container):

```
Regina 7.4.1  (Python 3.12.3)
  [PASS] H1(L(7,2)) = Z_7                                0.2 ms  Z_7
  [PASS] H1(Poincare) = 0                                0.1 ms  0
  [PASS] S3 recognised                                   0.1 ms  True
  [PASS] Poincare is not S3                             12.1 ms  False
  [PASS] RP3 # RP3 has 2 summands                        0.6 ms  2
  [PASS] census lookup: figure-eight = m004              2.1 ms  ['m004 : #1', 'L104001']
  [PASS] vol(figure-eight) ~ 2.029883                    0.1 ms  2.029883212819307
  [PASS] vol(Weeks) ~ 0.942707                           0.2 ms  0.9427073627769281
  [PASS] fig-8 has strict angle structure                0.1 ms  True
  [PASS] fig-8: 4 vertex quad normal surfaces            0.1 ms  4
  [PASS] TV_5 distinguishes S3 from L(5,1)               1.1 ms  ('1/10 x^3 - 1/10 x^2 +
1/5', '1/2')
  [PASS] Jones(right trefoil)                            0.1 ms  -x^8 + x^6 + x^2
  [PASS] fig-8 knot complement = m004                    1.9 ms  ['m004 : #1', 'L104001']
  [PASS] chi(CP2) = 3                                    0.2 ms  3
  [PASS] K3 intersection form: rank 22, sig -16, even  143.0 ms  (22, -16, True)
OK
```

The census lookup in checks 6 and 13 returns two hits, `m004` from the cusped hyperbolic census and `L104001` from the link census. Both name the figure-eight knot complement.

Environment notes

- The apt package builds `regina` for the distribution's system Python (3.12 on Ubuntu 24.04). In this container `python3` was 3.13 and could not import it. `scripts/find-python.sh` finds a working interpreter, and the Makefile uses it.
- Regina's `jones()` returns a Laurent polynomial in $x = \sqrt{t}$ (check 12). Keep this in mind when comparing with tables written in t.

Reproduce

```
scripts/install-regina-ubuntu.sh
make smoke        # or: make test (unit tests + smoke)
```